

Chapter 2 - The Foundational Operators of Universal Harmonic Theory

Zero Noise Equilibrium (ZNE) and Zero Node Yield (ZNY)

This chapter presents ZNE and ZNY as proposed foundational concepts within Universal Harmonic Theory. Established transport equations are distinguished from the proposed theoretical framework.

$$\begin{aligned}C(r, t) &= C_{eq}(r) + \delta C(r, t) \\ \delta C &= C - C_{eq} \\ Z_{NE}: \delta C &\rightarrow 0 \\ \lim_{\lambda \rightarrow 0} \{E(\lambda)\} &= E_{ideal} \\ Z_{NY} &= N_{coherent} / N_{total}, \quad 0 \leq Z_{NY} \leq 1 \\ \partial C / \partial t &= -\nabla \cdot J + S \\ J &= -D \nabla C + C v \\ J_{CG} &= Z_{NE} Z_{NY} (-D_G \nabla C + C v_G) \\ H_S &= Z_{NE} Z_{NY} B_{CG}(C, G)\end{aligned}$$

Foundational Postulate: Universal Harmonic Theory proposes that systems first approach equilibrium through ZNE and are then characterised by ZNY as a measure of coherent harmonic organisation.

Scientific Status: ZNE and ZNY are proposed theoretical constructs requiring validation.